FIRST ORDER THICKENINGS ARE EXACTLY SQUARE-ZERO QUASI-COHERENT IDEAL EXTENSIONS OF THE STRUCTURE SHEAF

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-more-morphisms/retained-preamble.tex), a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros; the Xy-pic xy environment is required because the counted proof displays two commutative diagrams. No macro was invented or redefined.

Editorial note. The two cross-chapter standard prerequisites are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- Thickening and first order thickening of schemes over a base. Same chapter, Definition environment labelled definition-thickening (source line 36); the statement refers to 'a first order thickening over S' and to $X' = (X, A)$. Supplies the definition that the counted proof has to verify for the locally ringed space (X, A); the verification itself stays inside the counted span.
- Cohomology of a quasi-coherent sheaf on an affine scheme vanishes in positive degrees. Cohomology of Schemes, Lemma labelled coherent-lemma-quasi-coherent-affine-cohomology-zero (canonical Stacks tag 01XB); invoked at proof lines 114-115. Supplies H^1(U, I) = 0, which is what makes A -> B = Gamma(U, O_X) surjective on an affine open; the rest of the argument is retained in the counted span.
- Morphisms from a locally ringed space to an affine scheme. Schemes, Lemma labelled schemes-lemma-morphism-into-affine (canonical Stacks tag 01I1); invoked at proof lines 118-120. Supplies the canonical morphism (U, A|_U) -> Spec(A) coming from the ring map B -> Gamma(U, A); the comparison of local rings is retained in the counted span.

Source reference. Original lemma 01XB: sources/stacks-chow/tags.txt, source lines 2512–2512. Cross-chapter standard prerequisite invoked once at proof lines 114-115 for the vanishing H^1(U, I) = 0 on an affine open; it supplies that vanishing only, while the local-ring comparison that concludes the proof is retained in the counted span.

Source reference. Original lemma 01I1: sources/stacks-chow/tags.txt, source lines 1962–1962. Cross-chapter standard prerequisite invoked once at proof line 119 for the existence of the canonical morphism (U, A|_U) -> Spec(A); the two commutative diagrams and the local-ring conclusion are retained in the counted span.

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *Let X be a scheme over a base S. Consider a short exact sequence*

$$0 \to \mathcal{I} \to \mathcal{A} \to \mathcal{O}_X \to 0$$

of sheaves on X where $\mathcal{A}$ is a sheaf of $f^{-1}\mathcal{O}_S$-algebras, $\mathcal{A} \to \mathcal{O}_X$ is a surjection of sheaves of $f^{-1}\mathcal{O}_S$-algebras, and $\mathcal{I}$ is its kernel. If

(1) *$\mathcal{I}$ is an ideal of square zero in $\mathcal{A}$, and*
(2) *$\mathcal{I}$ is quasi-coherent as an $\mathcal{O}_X$-module*

then $X' = (X, \mathcal{A})$ is a scheme and $X \to X'$ is a first order thickening over S. Moreover, any first order thickening over S is of this form.

Proof. It is clear that X' is a locally ringed space. Let $U = \mathrm{Spec}(B)$ be an affine open of X. Set $A = \Gamma(U, \mathcal{A})$. Note that since $H^1(U, \mathcal{I}) = 0$ (see Cohomology of Schemes, Lemma 01XB) the map $A \to B$ is surjective. By assumption the kernel $I = \mathcal{I}(U)$ is an ideal of square zero in the ring A. By Schemes, Lemma 01I1 there is a canonical morphism of locally ringed spaces

$$(U, \mathcal{A}|_U) \longrightarrow \mathrm{Spec}(A)$$

coming from the map $B \to \Gamma(U, \mathcal{A})$. Since this morphism fits into the commutative diagram

$$\begin{array}{ccc} (U, \mathcal{O}_X|_U) & \longrightarrow & \mathrm{Spec}(B) \\ \downarrow & & \downarrow \\ (U, \mathcal{A}|_U) & \longrightarrow & \mathrm{Spec}(A) \end{array}$$

we see that it is a homeomorphism on underlying topological spaces. Thus to see that it is an isomorphism, it suffices to check it induces an isomorphism on the local rings. For $u \in U$ corresponding to the prime $\mathfrak{p} \subset A$ we obtain a commutative diagram of short exact sequences

$$\begin{array}{ccccccccc} 0 & \longrightarrow & I_\mathfrak{p} & \longrightarrow & A_\mathfrak{p} & \longrightarrow & B_\mathfrak{p} & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & \mathcal{I}_u & \longrightarrow & \mathcal{A}_u & \longrightarrow & \mathcal{O}_{X,u} & \longrightarrow & 0. \end{array}$$

The left and right vertical arrows are isomorphisms because $\mathcal{I}$ and $\mathcal{O}_X$ are quasi-coherent sheaves. Hence also the middle map is an isomorphism. Hence every point of $X' = (X, \mathcal{A})$ has an affine neighbourhood and X' is a scheme as desired. □